Swamy B S

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Education

R V College of Engineering Sept 2023 - Jun 2027
B.E. in Information Science and Engineering — CGPA: 8.48/10 *Bengaluru, Karnataka*

GIPUC Channarayapattana Mar 2021 - Jun 2023
Pre-University — Percentage: 94.6% *Hassan, Karnataka*

Projects

CloudOps - Automated Deployment Platform for AWS Jun 2026 [GitHub]
Tech Stacks: React.js, Node.js, MongoDB, Docker, AWS
- Constructed a **self-service deployment platform** that automates a **4**-step pipeline: GitHub repository cloning, **Docker** image building, image publishing to **Amazon ECR**, and application deployment on **AWS EC2**.
- Automated **Docker** image generation with production-ready configurations, including **3** production safeguards: non-root execution, health checks, and graceful shutdown support.
- Integrated **Amazon ECR** with **AWS EC2** to automate **container deployment** and application health validation.
- Configured **GitHub Webhooks** for automated deployments and streamed **real-time deployment logs** using **Socket.io** while securely masking sensitive credentials.

PeerShare - Real-Time WebRTC P2P Communication Platform [GitHub] [Live]
Tech Stacks: WebRTC, Node.js, Socket.io, React.js, JavaScript
- Established a browser-based **peer-to-peer file sharing** and **video calling** application using **WebRTC**, enabling direct communication between connected peers without centralized storage.
- Developed a **Node.js** signaling server using **Socket.io** for peer discovery, SDP/ICE candidate exchange, connection establishment, and session management with STUN/TURN support.
- Built **chunked, resumable file transfers** with **SHA-256** integrity verification and adaptive flow control to improve reliability during large-file transfers.
- Optimized data processing using **Web Workers** and throughput monitoring, with automatic reconnection and resume logic for interrupted transfers.

KCET EduGuide - KCET Counselling & College Recommendation Platform [GitHub] [Live]
Tech Stacks: React.js, Node.js, Express.js, MongoDB, JWT, Recharts, Tailwind CSS, AWS EC2, AWS S3, Vercel
- Designed a **full-stack** counselling platform used by **70+** students to prepare KCET choice lists through personalized, **rank-based college and branch recommendations**.
- Crafted a responsive **React** frontend with interactive cutoff visualizations (**Recharts**), **PDF choice list generation**, and advanced filtering across **5** criteria: college, branch, category, location, and counselling round.
- Implemented **Node.js** and **Express.js REST APIs** with **MongoDB** covering **4** core areas: **recommendation algorithms**, **JWT authentication**, mentor verification, and meeting management.
- Deployed the backend on **AWS EC2** and the frontend on **Vercel**, using **AWS S3** to host images and branch insight JSON files.

Technical Skills

Languages: C, C++, JavaScript, SQL

Frontend: React.js, HTML, CSS, Vite, Tailwind CSS, Recharts

Backend: Node.js, Express.js, REST APIs, Socket.io, WebRTC, JWT Authentication

Databases: PostgreSQL, MongoDB, Redis

Cloud & DevOps: AWS (EC2, ECR, S3), Docker, Vercel

Tools: Git, GitHub, GitHub Webhooks, Linux, Postman

Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP

Achievements

Runner-Up - Coding Chef, Nagarjuna College of Engineering (May 2026) [Certificate]

Runner-Up - Coding Competition, KLE Institutions (Sept 2025) [Certificate]